

AI Marketing Compliance Review Reliability

Independent AI Systems Case Study

Problem

A simulated SaaS client needed a review workflow for marketing copy. The system had to reject explicit prohibited claims, route review worthy claims to humans, allow ordinary product descriptions, preserve every simultaneous issue, and ignore instructions embedded inside untrusted marketing text.

Frozen rule families

RULE 1 prohibited claims. RULE 2 measurable outcomes. RULE 3 absolute superiority. RULE 4 negative named competitor claims. RULE 5 unclear customer statement attribution. RULE 6 ordinary product descriptions. RULE 7 untrusted instructions inside source content.

Design goal

Use the smallest reliable architecture. Put semantic interpretation in the language model. Put exact lexical scanning, validation, and final business decisions in deterministic software. Preserve genuine uncertainty instead of forcing a false binary decision.

Architecture Boundary

Stage	Primary responsibility
1. Deterministic preprocessing	Preserve original copy and supply exact RULE 1 phrase candidates
2. Semantic reviewer	Interpret context, negation, sarcasm, claims, attribution, uncertainty, and evidence
3. Deterministic validation	Parse JSON, validate schema, allowed values, exact evidence, duplicates, candidate eligibility
4. Deterministic decision	Map issue states to APPROVED, REVIEW, or REJECTED
5. Human review	Resolve ambiguity and claims requiring policy or evidence judgment

Key restraint

The model never calculates the final submission outcome. RULE 1 with CONFIRMED maps to REJECTED. RULE 1 with UNCERTAIN maps to REVIEW. RULE 2 through RULE 5 map to REVIEW. No issues maps to APPROVED.

Development Benchmark V1

The development benchmark contained 31 synthetic English marketing submissions. Expected answers were frozen before execution. Because the same Veteran Prompt Engineering AI conversation had already discussed the benchmark, this run is described as a controlled same conversation development test, not an unseen evaluation.

31 of 31	Expected semantic submission decisions
24 of 24	Expected individual issues
27 of 31	Raw responses valid JSON
0 observed	High risk false approvals in this controlled set

What the benchmark attacked

Clear prohibited claims, discussion versus assertion, negation, measurable claims with and without digits, superiority, competitor criticism, customer attribution, simultaneous issues, sarcasm, ambiguity, long surrounding text, embedded AI instructions, lexical boundaries, exact whitespace preservation, and multiple occurrences of the same rule.

Unseen Synthetic Holdout

A new 10 submission holdout was created outside the Veteran AI conversation. The new submissions and expected answers were not supplied to the Veteran AI before execution.

10 of 10	Expected semantic submission decisions
9 of 9	Expected individual issues
9 of 9	Expected evidence selections
9 of 10	Raw responses valid JSON
0 observed	High risk false approvals in this small holdout

Observed difficult behavior

The reviewer correctly separated harmless policy discussion from prohibited marketing assertions, recognized a non digit measurable claim, detected superiority and competitor criticism, handled customer attribution, ignored an embedded instruction while preserving genuine violations, and preserved a genuine RULE 1 uncertainty case as REVIEW rather than forcing rejection or approval.

The Important Failure

Semantic correctness did not guarantee valid machine readable output.

Development raw JSON	27 of 31 valid
Holdout raw JSON	9 of 10 valid
Combined raw JSON	36 of 41 valid
Combined invalid raw responses	5 of 41

Failure pattern

The recurring failure involved quotation marks inside exact source evidence that were not escaped correctly. The semantic rule, issue state, and evidence meaning were still correct, but a strict parser would reject the raw response.

Architecture conclusion

Do not keep adding instructions such as remember to escape quotes. JSON parsing, schema enforcement, allowed enums, exact evidence verification, duplicate handling, final outcome mapping, and malformed output rejection belong in deterministic infrastructure or schema constrained generation.

Responsibility Split

Language model

Interpret context, negation, quotation, sarcasm, customer attribution, measurable claims, superiority claims, competitor criticism, genuine semantic uncertainty, and the smallest supporting evidence span.

Deterministic software

Perform RULE 1 lexical scanning, JSON parsing, schema validation, allowed value enforcement, exact source substring verification, duplicate handling, rule to outcome mapping, and rejection or retry of malformed responses.

Human review

Resolve genuine ambiguity, evaluate external evidence, make policy judgments that are not fully encoded, handle competitor and testimonial risk decisions, and review cases that automated validation cannot safely repair.

Portfolio Conclusion

Case Study 3 demonstrates a reusable AI reliability method in a third business domain. Case Study 1 focused on classification, overlapping labels, and partial uncertainty. Case Study 2 focused on evidence grounded extraction, provenance, OCR uncertainty, and conflicting source material. Case Study 3 focuses on rule based marketing review, deterministic lexical controls, semantic interpretation, exact source evidence, uncertainty, prompt injection resistance inside untrusted content, and deterministic business outcome mapping.

Truthful conclusion

Across the controlled development benchmark and small unseen synthetic holdout, the frozen semantic reviewer matched the expected decisions for all 41 submissions. This does not establish production accuracy. The experiment also exposed five raw JSON failures, showing why semantic reasoning and deterministic machine guarantees must remain separate.

Limitations

Synthetic examples only. Conversational testing rather than independent API execution. Small benchmark sizes. No production traffic. No cost, latency, retries, throughput, privacy, monitoring, or deployment measurements. No externally validated compliance performance. Controlled prompt injection tests are not a security proof.

Independent AI systems case study. Results reflect controlled synthetic testing only.